

LESSON 5.5

SCALE FACTORS - PERIMETER



LESSON INTRODUCTION

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

LEARNING TARGETS

- I can relate the perimeters of rectangles to their scale factors.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

ESSENTIAL QUESTIONS

- What are scale models, and how are they used?
- What is the relationship between an original figure's perimeter and its scale model's perimeter?
- How does size of the scale factor relate to the size of the scale model?

LESSON NARRATIVE

This lesson activates prior knowledge determining the perimeter of figures and extends it to the idea of scale models. Students begin in the 5.5.1 Warm-Up, looking at scale factors as ratios without the terminology. As the lesson progresses, students are given the necessary vocabulary to extend this understanding to determining the scale factors relating pairs of figures and exploring how these scale factors relate to the relationship between the perimeters of the figures.

COMPONENT SUMMARY

5.5.1 WARM-UP (~5 MIN.)

Find the dimensions of two rectangles so that the dimensions form equivalent ratios

5.5.3 EXPLORATION (10-15 MIN.)

Explore how the perimeters of similar figures are related to the scale factor

5.5.2 GUIDED INSTRUCTION (15-20 MIN.)

Determine what a scale model is, what corresponding parts are, and what a scale factor is

5.5.4 WRAP-UP (~5 MIN.)

Determine the scale factor of a scale drawing and relate the perimeters using the scale factor

RESOURCES

GLOSSARY

Key Terms:

- Scale model
- Corresponding parts
- Scale factor

Additional Terms:

- N/A

MATERIALS

- (Optional) Graph paper
- (Optional) Crayons, colored pencils, or highlighters

ADDITIONAL PREPARATION

- This is a vocabulary-rich lesson. Consider preparing your classroom-specific vocabulary or glossary strategies for support (word wall, math cards, etc.)

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may incorrectly set up the ratio to find the perimeter of a similar figure, either reversing the ratio or writing the ratio with noncorresponding sides.
- Students may not connect scale factor with multiplication when the scale factor is a fraction, instead thinking it must be division.

THINGS TO CONSIDER

- Corresponding sides may not be easy for everyone to see. Color-coding may be a helpful tool.
- Considering how the sizes of the figures relate to one another first allows students to check the reasonableness of their answers when determining the scale factor as they develop the understanding that scale factors greater than 1 enlarge the figure and scale factors between 0 and 1 reduce the figure.
- Ensure students are connecting scale factor with multiplication either by a factor greater than 1 or between 0 and 1.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

5.5.2 Guided Instruction includes a question where students discuss with their partners what a scale model is. This provides an opportunity for students to engage in mathematical discourse and solidify understanding through partner discussion.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

Component 5.5.3 is an Exploration in which students are given multiple examples of a figure and its scaled model. They use the concepts presented to them in this component to look for patterns as they determine the relationship between the scale factor relating to figures and the ratio of the perimeters of those figures.

DIFFERENTIATE INSTRUCTION

This lesson provides multiple opportunities to differentiate content for various learning preferences. Corresponding sides may be not be easy for everyone to see. Color-coding may be a helpful tool and provides a visual entry point. Encourage students to use colored pencils or highlighters to make the sides that correspond. This also provides a quick check while circulating to ensure students are relating the figures appropriately.

Additional differentiation opportunities are denoted throughout the lesson guide.

5.5.1 WARM-UP: SCALED COPIES

ENRICHMENT

Students have worked with equivalent ratios. While they have not explicitly learned about scale drawings, the scaffolding in the question stem allows all students the opportunity to access the question. Resist the urge to guide students' thinking on this Warm-Up, as the key takeaways will be discussed in the next component. Instead, allow students a few minutes to explore the question and see what they are able to come up with based on what they know and the information given.

There are multiple correct answers to this question using a variety of scale factors. If many students complete the Warm-Up on their own, consider facilitating a brief share of the dimensions they came up with.

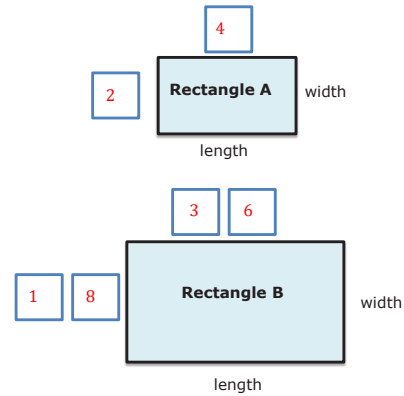
Other possible answers include:

- A: 1×4 and B: 03×12
- A: 2×5 and B: 16×40



5.5.1 Warm-Up: Scaled Copies

Two rectangles are shown with their dimensions missing.



1. What are the lengths and widths of Rectangle A and Rectangle B that make the ratios $\frac{\text{width of Rectangle A}}{\text{width of Rectangle B}}$ and $\frac{\text{length of Rectangle A}}{\text{length of Rectangle B}}$ equal? Fill in the blue boxes using only the digits 0 – 9 so that each box has a different number inside.

Answers will vary. One example shown above.



5.5.2 Guided Instruction: Scale Drawings and Scale Factors

In the warm-up, the ratio of the widths of rectangles A and B is equivalent to the ratio of the lengths of rectangles A and B. Therefore, rectangle A is a scale model of rectangle B.



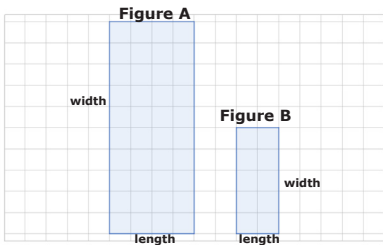
A **scale model** is a model or drawing based on a ratio of the dimensions for the model and the actual object it represents.

1. The keychain shown of the Eiffel Tower is a scale model of the real Eiffel Tower in Paris.



- a. Sketch a drawing of the Eiffel Tower that is not a scale model.
Answers will vary. Sketch should not be proportional to the Eiffel Tower.
- b. Discuss with your partner why your drawing is not a scale model. Write a summary of your discussion.
This drawing is not a scale model because the drawing isn't based on a ratio of the actual dimensions of the actual Eiffel Tower. I drew long legs and a short tower. I do not know this for certain, because I did not use the actual measurements of the Eiffel Tower when conducting my drawing, but a glance of the sketch shows me that I do not have a scale model.

2. Celeste drew an outline of a building for a diagram, and then drew a second model slightly smaller than her original drawing. Each gridline represents 1 centimeter (cm).



The length of Figure A and the length of Figure B can be written as a ratio because they are corresponding parts. This is also true for the widths of the figures.

Corresponding parts are parts of each figure that are in the same position in relation to the rest of each figure.

3. Complete the table for Figures A and B. Each gridline represents 1 cm.

	Length (cm)	Width (cm)
Figure A	4	10
Figure B	2	5
Ratio of Corresponding Parts Figure A Figure B	$\frac{4}{2}$	$\frac{10}{5}$

5.5.2 GUIDED INSTRUCTION: SCALE DRAWINGS AND SCALE FACTORS

DIFFERENTIATE INSTRUCTION

There are three major vocabulary terms that students need to continue working with scale models. Consider having students create a Frayer Model for each term to encourage synthesis. Review the definition of each term as a class, then have students come up with examples, nonexamples, and characteristics.

TEACHER MOVES

Think-Pair-Share

Engage students in a Think-Pair-Share routine as they discuss question 1b. This gives students the opportunity to consider what makes a model a scale model and verbalize their reasoning before putting it in writing. It encourages mathematical discourse, which deepens understanding as students revise their responses and build upon each other's thoughts.

ENRICHMENT

Does it matter if we read this ratio as 2 to 4 or 4 to 2? Why or why not?

TEACHER MOVES

Students can utilize the glossary video for additional support in understanding "corresponding parts."

5.5.2 GUIDED INSTRUCTION: SCALE DRAWINGS AND SCALE FACTORS (CONTINUED)

TEACHER MOVES

Guide student discussion by asking the following questions:

- What do you notice about the scale factor when the figure is enlarged?
- What do you notice about the scale factor when the figure is reduced in size?

A model is a scale model if the ratios of the corresponding parts of the figures are equivalent. These ratios can be used to determine the scale factor that relates the figures.



A **scale factor** is the constant that is multiplied by the length of each side of a figure to produce an image that is the same shape as the original figure.

4. Discuss with your partner what you notice about the scale factor that uses $\frac{\text{Figure A}}{\text{Figure B}}$ and the scale factor that uses $\frac{\text{Figure B}}{\text{Figure A}}$. Write a summary of your discussion.

Answers will vary. My partner and I noticed the ratio of the dimensions of Figure A to Figure B is 2:1, and the ratio of the dimensions of Figure B to Figure A is 1:2.

Scale models can be scaled up or scaled down.

Scaled Down From the Original	Original	Scaled Up From the Original
		

TEACHER MOVES

Consider giving the students time to collaborate with their partner to complete the statements before going over the correct responses. Circulate while students are discussing with their partner to determine whether any clarifications need to be made before students move on to the 5.5.3 Exploration.

5. Complete the statements about Figure A and Figure B.
The scale factor that relates Figure A to Figure B is $\frac{1}{2}$ because each side length of Figure A is multiplied by this scale factor to get the side lengths of Figure B.

This is an example of scaling a figure ☐ up ☐ down because the figure gets ☐ larger. ☐ smaller.

The scale factor that relates Figure B to Figure A is $\frac{2}{1}$, which is an example of scaling the figure

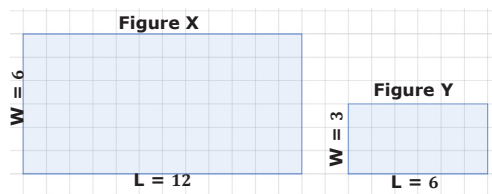
☐ up ☐ down because the figure gets ☐ larger. ☐ smaller.



5.5.3 Exploration: Scale Factor and Perimeter

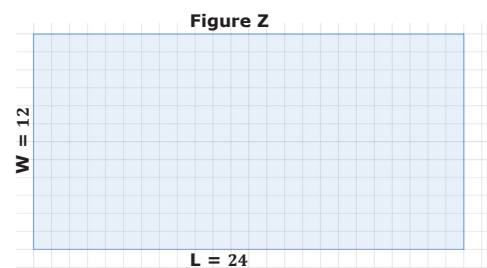
Work with your partner to complete the following.

- Paolo drew Figure X to represent a picture he's putting in an advertisement he is designing. Paolo's rectangle has a length (L) of 12 units by a width (W) of 6 units. Because he has several design options for his advertisement, he also drew a scale model, Figure Y, with a length (L) of 6 units and a width (W) of 3 units.



- The scale factor from Figure X to Figure Y is $\frac{1}{2}$.
- What is the perimeter of Figure X?
 $12 + 12 + 6 + 6 = 36 \text{ units}$
- What is the perimeter of Figure Y?
 $3 + 3 + 6 + 6 = 18 \text{ units}$
- What is the ratio of the perimeter of Figure Y to the perimeter of Figure X?
 $18:36 \text{ which reduces to } 1:2, \text{ or } \frac{1}{2}$

Paolo decided to change the scale factor to 2 to create Figure Z from Figure X. Figure Z has a length (L) of 24 units by a width (W) of 12 units.



- What is the perimeter of Figure Z?
 $12 + 12 + 24 + 24 = 72 \text{ units}$
- How does that compare to the perimeter of Figure X?
The perimeter of Figure Z is 72 units, which is 2 times the perimeter of Figure X, 36 units. The ratio of the perimeter of Figure Z to Figure X is 2:1, or $\frac{2}{1}$.
- What relationship, if any, is there between the scale factor and the ratio of the perimeters?
Answers will vary. The scale factor used to relate the dimensions of the figures can also be used to relate the perimeters. The ratio of the perimeters is equal to the scale factor.

5.5.3 EXPLORATION: SCALE FACTOR AND PERIMETER

QUESTIONING

Consider a billboard extension project in which students design their own billboard using Paolo's scale from question 1. Provide students with the exact dimensions of a billboard, and for homework, students must identify the scale factor, create their own billboard, and use that scale factor to identify the measurements of images within their billboard.

ENRICHMENT

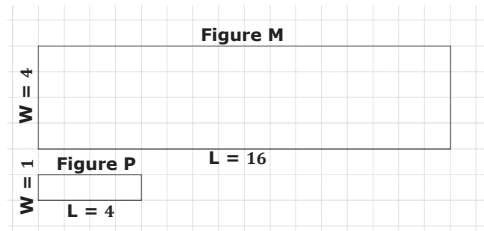
- What conclusion can we draw about the perimeter of Figure Z compared to Figure X? (The perimeter of Figure Z is twice the length of the perimeter of Figure X.)
- What conclusion can we draw about the perimeter of Figure X compared to Figure Z? (The perimeter of Figure X is half the length of the perimeter of Figure Z.)
- Can more than one scale factor be used to describe the relationship between two figures?
- Does the order in which the figures are related change the scale factor used?

5.5.3 EXPLORATION: SCALE FACTOR AND PERIMETER (CONTINUED)

QUESTIONING

For students who finish early, consider asking them to design their own polygon. Then from their original, create a scaled copy. Students may choose to scale their figure up or down.

2. Figure M has a length (L) of 16 units and a width (W) of 4 units. Figure P is a scale model of Figure M. Figure P has a length (L) of 4 units and a width (W) of 1 unit.

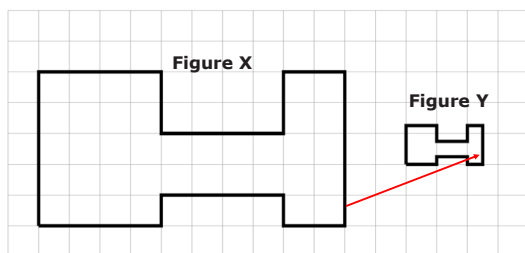


- The scale factor from Figure M to Figure P is $\frac{1}{4}$.
 - What is the perimeter of Figure M?
 $4 + 4 + 16 + 16 = 40 \text{ units}$
 - Without finding the sum of the sides, what do you expect the perimeter of Figure P to be?
 $40 \times \frac{1}{4} = 10 \text{ units}$
 - Verify your answer by finding the sum of the sides of Figure P.
 $1 + 1 + 4 + 4 = 10 \text{ units}$
3. With your partner, write a rule that can be used to calculate the perimeter of a scale model if given only an original perimeter and a scale factor.
Answers will vary. You can find the unknown perimeter of a scale model by multiplying the perimeter of one figure by the scale factor used to scale to the other figure.



5.5.4 Wrap-Up: Ticket Out the Door

- The second H-shaped polygon, Figure Y, is a scaled copy of the first, Figure X.



- Show one pair of corresponding sides in the original polygon and its copy by drawing an arrow from Figure X to Figure Y.

Answers will vary. Example shown on model.

- What scale factor takes the original H-shaped polygon, Figure X, to its smaller copy, Figure Y?

The scale factor from Figure X to Figure Y is $\frac{1}{4}$.

- What is the ratio of the perimeter of Figure Y to the perimeter of Figure X?

1:4, or $\frac{1}{4}$

If you work out the sides the perimeters are 36 units and 9 units.

5.5.4 WRAP-UP: TICKET OUT THE DOOR

QUESTIONING

Prompt students who are ready for a challenge to determine if the ratio for the area would be 4:1. Ask them to show their reasoning using math or a model.